

Lab program-25:

25. Construct a C program to implement the I/O system calls of UNIX (fcntl, seek, stat, opendir, readdir).

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>
#include <dirent.h>

int main(void) {
    const char *filename = "foo.txt";
    int fd = open(filename, O_RDWR | O_CREAT, 0644);
    if (fd == -1) {
        perror("open");
        return 1;
    }

    printf("File descriptor: %d\n", fd);

    int flags = fcntl(fd, F_GETFL);
    if (flags == -1) {
        perror("fcntl");
        close(fd);
        return 1;
    }
    printf("File access flags obtained using fcntl.\n");
}
```

```
const char *text = "UNIX I/O system calls example.\n";
if (write(fd, text, 31) == -1) {
    perror("write");
    close(fd);
    return 1;
}

if (lseek(fd, 0, SEEK_SET) == -1) {
    perror("lseek");
    close(fd);
    return 1;
}
printf("File pointer moved using lseek.\n");

struct stat st;
if (stat(filename, &st) == -1) {
    perror("stat");
    close(fd);
    return 1;
}
printf("File size: %ld bytes\n", (long)st.st_size);
close(fd);

DIR *dir = opendir(".");
if (dir == NULL) {
```

```
        perror("opendir");
        return 1;
    }

    printf("Directory entries:\n");
    struct dirent *entry;
    while ((entry = readdir(dir)) != NULL)
        printf("%s\n", entry->d_name);

    closedir(dir);
    return 0;
}
```

OUTPUT:

```
fd = 3

-----
Process exited after 0.08362 seconds with return value 0
Press any key to continue . . .
```